

Dataset Summary Report

SentinAI: National Crime Analytics and Predictive Policing Dataset

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1. Executive Summary

This report presents the dataset summary for the SentinAI crime analytics project. The project uses a cleaned, harmonized, and demographically enriched crime dataset for India to support exploratory crime analysis, hotspot forecasting, incident category prediction, and responsible AI-based decision support.

Two major datasets are used:

Dataset	File Name	Records	Columns	Coverage
District-year crime dataset	india_crime_master_dataset.csv	22,243	43	2001-2026
Incident-level crime dataset	india_crime_incidents_master.csv	380,145	13	2020-2026

The district-year dataset combines crime counts with demographic and geographic indicators such as population, literacy rate, sex ratio, SC/ST percentage, urbanization, district area, latitude, longitude, population density, and crime per capita. The incident-level dataset provides fine-grained simulated FIR-style records with timestamps, police station attribution, location coordinates, and crime categories.

The dataset has been prepared to support both research and dashboard implementation. It enables trend analysis, geospatial hotspot visualization, machine learning model training, and ethical review of predictive policing risks.

2. Purpose of the Dataset

The purpose of preparing this dataset is to build a reliable analytical foundation for SentinAI, an AI/ML-based crime analytics system. The dataset supports the following objectives:

1. Analyze historical crime trends across Indian states, union territories, and districts.
2. Identify high-risk crime hotspots using crime-per-capita indicators.
3. Train machine learning models for annual district hotspot forecasting.
4. Train crime category prediction models for incident-level decision support.
5. Support interactive dashboard visualizations using maps, filters, charts, and prediction panels.
6. Study fairness and ethical concerns related to predictive policing.

3. Dataset Overview

3.1 District-Year Master Dataset

The main dataset, `india_crime_master_dataset.csv`, contains district-level crime statistics and demographic indicators.

Attribute	Value
Total records	22,243
Total columns	43
Time period	2001 to 2026
Number of years	26
States/UTs represented	36
Unique districts represented	944
Total IPC crime records summed across dataset	77,145,905
Average crime per capita	175.39 per 100,000 population
Median crime per capita	138.33 per 100,000 population
75th percentile hotspot threshold	216.78 per 100,000 population

This dataset is suitable for longitudinal crime analysis because each row represents a district-year observation. It contains both crime counts and contextual demographic features.

3.2 Incident-Level Dataset

The second dataset, `india_crime_incidents_master.csv`, provides incident-level simulated

FIR-style logs for detailed spatiotemporal analysis.

Attribute	Value
Total records	380,145
Total columns	13
Time period	2020 to 2026
Number of years	7
States/UTs represented	36
Districts represented	725

This dataset supports map-based hotspot visualization and crime category prediction. It includes timestamp, police station, crime category, latitude, longitude, and demographic context.

4. Data Sources

The project dataset was prepared by combining and harmonizing multiple official and public data sources:

- National Crime Records Bureau (NCRB) reports, 2001-2012**
Historical district-wise crime data from "Crime in India" reports.
- Open Government Data Platform India, 2014**
Crime data published through the Government of India's open data portal.
- NCRB district-wise crime reports, 2024**
Official 2024 district-level crime data integrated from Excel sheets.
- Census 2011 demographic information**
Population, literacy rate, sex ratio, area, SC/ST percentage, urbanization, and related district indicators.
- Geospatial district centroids**
Latitude and longitude values used for map visualization and spatial modeling.

The dataset includes official reported crime records wherever available. Missing years and incident-level records were generated through controlled projection and simulation methods, described below.

5. Data Cleaning and Harmonization

5. Data Cleaning and Harmonization

Preparing the dataset required several cleaning and standardization steps:

5.1 Column Standardization

Different data sources used different column names and crime category structures. These were mapped into a common schema so that crime categories could be compared across years.

Examples:

Source Variation	Standardized Field
Hurt and grievous hurt categories	HURT/GREVIOUS HURT
Motor vehicle theft	AUTO THEFT
Other theft categories	OTHER THEFT
Total cognizable IPC crimes	TOTAL IPC CRIMES

5.2 Removal of Aggregate Rows

Raw NCRB datasets often include district rows, state total rows, and national total rows. State-level and national-level total rows were removed to prevent double-counting.

Examples of removed aggregate-style rows:

Removed Row Type	Reason
TOTAL	Not an individual district
ALL INDIA	National aggregate
State total rows	Would duplicate district-level totals

5.3 Geographic Standardization

State and district names were cleaned by:

1. Removing extra spaces.
2. Standardizing uppercase names.
3. Reconciling spelling variations.
4. Mapping changed or split districts to consistent administrative references.
5. Aligning district records with Census 2011 parent district boundaries where needed.

5.4 Missing Value Handling

Missing or non-numeric crime counts were converted to zero where appropriate. This ensured that the dataset remained complete and usable for model training and dashboard analysis.

5.5 Demographic Enrichment

Crime records were merged with demographic variables to create a richer dataset. The following attributes were added:

Variable	Description
POPULATION	District population
LITERACY_RATE	Literacy rate percentage
SEX_RATIO	Females per 1000 males
SC_ST_PERCENTAGE	Combined SC/ST population percentage
AREA_SQ_KM	District area in square kilometers
LATITUDE	District centroid latitude
LONGITUDE	District centroid longitude
URBAN_RATIO	Urbanization ratio
POPULATION_DENSITY	Population divided by area
CRIME_PER_CAPITA	Total IPC crimes per 100,000 population

6. Projection and Simulation Methodology

6.1 Projected Annual Data

To maintain a continuous longitudinal dataset from 2001 to 2026, projected records were created for years where official district-level continuity was incomplete. The projection approach used historical district-specific trends, controlled growth rates, and statistical noise to preserve realistic variation.

The projected years are useful for model experimentation and dashboard demonstration. However, they should be clearly identified as projected data and not treated as fully official NCRB records.

6.2 Incident-Level Simulation

The incident-level dataset was generated to resemble operational FIR-style logs. It was

created from the annual crime dataset using proportional sampling and realistic temporal-spatial patterns.

Simulation features include:

- 1. Unique incident IDs.
- 2. Crime category assignment.
- 3. District and police station attribution.
- 4. Timestamp generation.
- 5. Seasonal crime variation.
- 6. Hour-of-day crime patterns.
- 7. Latitude and longitude jitter around district centroids.

This enables hotspot mapping, temporal analysis, and machine learning model development at a finer level than annual district totals.

7. Dataset Schema Summary

7.1 Main Crime Categories

The district-year dataset includes major IPC crime categories such as:

Crime Category
Murder
Attempt to murder
Culpable homicide not amounting to murder
Rape
Kidnapping and abduction
Dacoity
Robbery
Burglary
Theft
Auto theft
Riots
Criminal breach of trust
...

Cheating
Counterfeiting
Arson
Hurt/grievous hurt
Dowry deaths
Assault on women with intent to outrage modesty
Insult to modesty of women
Cruelty by husband or relatives
Causing death by negligence
Other IPC crimes
Total IPC crimes

7.2 Incident-Level Fields

The incident-level dataset contains:

Field	Description
INCIDENT_ID	Unique FIR-style incident ID
STATE/UT	State or union territory
DISTRICT	District name
POLICE_STATION	Assigned police station
CRIME_CATEGORY	Crime category
TIMESTAMP	Incident date and time
LATITUDE	Crime location latitude
LONGITUDE	Crime location longitude
POPULATION	District population
LITERACY_RATE	District literacy rate
SEX_RATIO	District sex ratio
SC_ST_PERCENTAGE	District SC/ST percentage
URBAN_RATIO	District urbanization ratio

8. Descriptive Data Summary

8.1 Top States/UTs by Total IPC Crimes

Rank	State/UT	Total IPC Crimes
1	Maharashtra	6,973,662
2	Madhya Pradesh	6,675,860
3	Uttar Pradesh	6,419,718
4	Rajasthan	5,802,093
5	Tamil Nadu	5,400,262
6	West Bengal	5,286,337
7	Andhra Pradesh	5,203,675
8	Karnataka	4,536,728
9	Delhi	4,489,101
10	Bihar	4,438,553

8.2 Top Districts by Total IPC Crimes

Rank	State/UT	District	Total IPC Crimes
1	Maharashtra	Mumbai Commr.	880,011
2	Karnataka	Bangalore Commr.	830,583
3	Maharashtra	Mumbai	681,522
4	Karnataka	Bengaluru City	567,259
5	Madhya Pradesh	Indore	536,833
6	Bihar	Patna	522,702
7	West Bengal	Kolkata	518,509
8	Gujarat	Ahmedabad Commr.	502,090
9	Delhi	West	497,068
10	Delhi	Outer	490,272

8.3 Most Frequent Crime Categories in the District-Year Dataset

Rank	Crime Category	Total Cases
1	Other IPC Crimes	26,544,953
2	Theft	11,433,134
3	Hurt/Grievous Hurt	7,609,393
4	Other Theft	6,954,002
5	Auto Theft	4,480,608
6	Causing Death by Negligence	3,182,431
7	Burglary	3,081,482
8	Cruelty by Husband or His Relatives	2,995,508
9	Cheating	2,700,296
10	Riots	1,917,932
11	Assault on Women with Intent to Outrage Her Modesty	1,728,963
12	Kidnapping and Abduction	1,611,703

8.4 Yearly Crime Trend

The total IPC crimes increased from 1,769,308 in 2001 to 4,145,259 in 2026 in the prepared dataset. The year 2024 shows a lower total of 2,585,487 because it uses integrated actual 2024 records that were aligned with the cleaned district structure.

Year	Total IPC Crimes
2001	1,769,308
2002	1,780,330
2003	1,716,120
2004	1,832,010
2005	1,822,602
2022	4,025,798
2023	4,062,519
2024	2,585,487
2025	4,114,349
2026	4,145,259

9. Incident-Level Dataset Findings

9.1 Top Incident Categories

Rank	Crime Category	Incident Count
1	Theft	101,557
2	Other IPC Crimes	72,737
3	Hurt/Grievous Hurt	37,522
4	Other Theft	35,290
5	Auto Theft	19,572
6	Burglary	16,649
7	Causing Death by Negligence	15,798
8	Cheating	15,530
9	Cruelty by Husband or His Relatives	12,878
10	Kidnapping and Abduction	11,912

9.2 Top Districts/Cities in Incident Dataset

Rank	State/UT	District	Incident Count
1	Maharashtra	Mumbai	21,489
2	Delhi	Outer	18,322
3	Karnataka	Bangalore Commr.	18,023
4	Delhi	West	16,792
5	Delhi	South	15,068
6	Delhi	East	14,336
7	Delhi	North-East	14,076
8	Delhi	South-East	12,594
9	Telangana	Cyberabad	9,759
10	Uttar Pradesh	Lucknow	9,450

9.3 Time-Based Incident Pattern

The incident-level dataset shows stronger crime concentration during late-night hours. High-count hours include:

Hour	Incident Count
00:00	20,775
01:00	21,060
02:00	20,581
03:00	21,086
22:00	20,800
23:00	20,693

This pattern supports the use of time-aware patrol planning and hotspot visualization.

9.4 Monthly Incident Pattern

The incident dataset shows higher counts in January, February, November, and December.

Month	Incident Count
January	35,165
February	35,126
November	35,000
December	34,998

This suggests seasonal variation in the simulated incident logs, especially for property-related crimes.

10. Machine Learning Use of Dataset

The dataset supports two major machine learning tasks.

10.1 Hotspot Forecasting

The district-year dataset is used to classify whether a district is high-risk or low-risk based on crime per capita.

Model	Accuracy	F1- Score
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Random Forest	95.31%	0.9078
XGBoost	95.18%	0.9058
Decision Tree	94.75%	0.8978
Logistic Regression	93.46%	0.8646

The best-performing model is Random Forest. The hotspot threshold is based on the 75th percentile of crime per capita, approximately 216.78 crimes per 100,000 population.

10.2 Crime Category Prediction

The incident-level dataset is used to classify incidents into broader crime categories.

Model	Accuracy	Weighted F1- Score
XGBoost	60.09%	0.5352
Random Forest	59.95%	0.5333
Decision Tree	58.85%	0.5310
Logistic Regression	55.61%	0.4755

The best-performing model is XGBoost. It performs well for majority classes such as property crimes, while minority crime categories require further improvement.

11. Strengths of the Dataset

1. **Large longitudinal coverage:** The dataset covers 26 years from 2001 to 2026.
2. **District-level granularity:** The dataset supports localized analysis rather than only state-level analysis.
3. **Demographic enrichment:** Crime records are connected with census and socio-demographic indicators.
4. **Geospatial readiness:** Latitude and longitude fields support map visualization and hotspot detection.
5. **Machine learning suitability:** The dataset includes features needed for supervised learning.
6. **Dashboard compatibility:** The structure is suitable for Streamlit, Folium, Plotly, and model inference workflows.
7. **Ethics-aware design:** The dataset documentation clearly separates reported crime, projected data, and simulated incident records.

12. Limitations and Ethical Considerations

The dataset is useful for academic research and prototype development, but the following limitations must be considered:

- 1. Reported crime is not equal to actual crime.**
The dataset is based on reported FIRs and registered crime records. Under-reporting may be significant in sensitive categories such as rape, dowry deaths, and domestic cruelty.
- 2. Projected records are not official records.**
Some years were generated through projection to create a continuous time series. These records should be used for modeling experiments, not for official crime reporting.
- 3. Incident-level logs are simulated.**
The incident dataset is designed to support product demonstration and machine learning development. It should not be presented as actual FIR microdata.
- 4. Administrative boundaries change over time.**
District splits, state reorganizations, and renamed regions can affect comparability across years.
- 5. Predictive policing can create feedback loops.**
If a model sends more patrols to historically over-policed areas, more incidents may be recorded there, reinforcing biased predictions.
- 6. Human oversight is necessary.**
Model outputs should be advisory and should not replace legal, ethical, or administrative decision-making.

13. Recommendations

1. Use the dataset primarily for academic research, visualization, and prototype demonstration.
2. Clearly label projected and simulated records in all presentations and reports.
3. Improve minority category prediction using class balancing, cost-sensitive learning, or additional verified data.
4. Validate future models with official updated NCRB records when available.
5. Include fairness checks before any operational use.
6. Use human-in-the-loop review for all hotspot and incident category predictions.
7. Avoid using the model for punitive deployment without ethical approval and legal review.

14. Conclusion

The SentinAI dataset provides a strong foundation for crime analytics, hotspot forecasting, and responsible AI research. It combines historical crime data, demographic indicators, geospatial coordinates, and simulated incident-level records into a structured analytical system.

The dataset is suitable for internship submission, academic explanation, dashboard demonstration, and machine learning experimentation. At the same time, it must be presented responsibly: the annual dataset contains both official and projected records, while the incident-level dataset is simulated for demonstration and modeling purposes.

Overall, the dataset demonstrates how crime analytics can be enhanced through data cleaning, demographic enrichment, geospatial modeling, and ethical AI safeguards.

15. Files Included in the Dataset Work

File	Purpose
india_crime_master_dataset.csv	Main district-year crime and demographic dataset
india_crime_incidents_master.csv	Incident-level simulated FIR-style dataset
dataset_sources.md	Dataset source and attribute documentation
train_hotspot_predictor.py	Hotspot forecasting model training script
train_crime_category_predictor.py	Crime category prediction model training script
app.py	Streamlit dashboard application
models/best_hotspot_model.pkl	Saved hotspot model
models/best_category_model.pkl	Saved category prediction model